Wenqian Ye

Contact Information

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Research Interests

My research focuses on robustness and trustworthiness in machine learning systems, both **theoretically** and **practically**. I have extensively worked on fundamental ML research, including out-of-distribution (OOD) generalization, bayesian machine learning, and spurious correlations in machine learning. More recently, my work has extended to AI Alignment, with a particular focus on **mechanistically** understanding and mitigating emergent misalignment in Large Language Models (i.e., scheming and reward hacking) and Agentic Systems (i.e., posterior collapse). I am also interested in how agents can self-evolve and discover new tasks to push the scientific boundaries of humanity. My research has been applied to various applications such as embodied AI, healthcare, and AI for science.

Education

2023/08 – Present **University of Virginia** Charlottesville, VA
Ph.D. in Computer Science
Advisor: Aidong Zhang

2020/08 – 2022/05 **Courant Institute, New York University** New York, NY
M.S. in Computer Science
Concentration: Machine Learning

2017/08 – 2020/05 **University of Illinois Urbana-Champaign** Urbana, IL
B.S. in Mathematics
Thesis Advisor: Sanjay Patel
Minors in Computer Science and Electrical Engineering

Publications

Note: † denotes Equal Contribution.

Peer-Reviewed Conference/Journal Publications

[1] **Wenqian Ye**, Bohan Liu, Guangtao Zheng, Di Wang, Xu Cao, Yunsheng Ma, Bolin Lai, James Rehg, and Aidong Zhang. MM-SpuBench: Towards Better Understanding of Spurious Biases in Multimodal LLMs. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (*KDD*), 2026.
Oral Presentation at NeurIPS RBFM Workshop.

[2] **Wenqian Ye**, Di Wang, Guangtao Zheng, Bohan Liu, and Aidong Zhang. SAGE:

Spuriousness-Aware Guided Prompt Exploration for Mitigating Multimodal Bias. In *AAAI Conference on Artificial Intelligence* (*AAAI*), 2026.

[3] **Wenqian Ye**, Guangtao Zheng, and Aidong Zhang. Rectifying Shortcut Behaviors in Preference-based Reward Learning. In *Advances in Neural Information Processing Systems* (*NeurIPS*), 2025.

[4] **Wenqian Ye**, Guangtao Zheng, and Aidong Zhang. Improving Group Robustness on Spurious Correlation via Evidential Alignment. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (*KDD*), 2025.
Best Paper Award (Research Track).

[5] **Wenqian Ye**, Luyang Jiang, Eric Xie, Guangtao Zheng, Yunsheng Ma, Xu Cao, Dongliang Guo, Daiqing Qi, Zeyu He, Yijun Tian, Megan Coffee, Zhe Zeng, Sheng Li, Ting-hao Kenneth Huang, Ziran Wang, James M. Rehg, Henry Kautz, and Aidong Zhang. The Clever Hans Mirage: A Comprehensive Survey on Spurious Correlations in Machine Learning. In *Transactions on Machine Learning Research* (*TMLR*), 2025.
Shorter version at ICML DMLR Workshop.

[6] Guangtao Zheng, **Wenqian Ye**, and Aidong Zhang. NeuronTune: Towards Self-Guided Spurious Bias Mitigation. In *International Conference on Machine Learning* (*ICML*), 2025.

[7] Guangtao Zheng, **Wenqian Ye**, and Aidong Zhang. ShortcutProbe: Probing Prediction Shortcuts for Learning Robust Models. In *International Joint Conference on Artificial Intelligence* (*IJCAI*), 2025.

[8] Wei Qian, Chenxu Zhao, Yangyi Li, **Wenqian Ye**, and Mengdi Huai. Towards Unveiling Predictive Uncertainty Vulnerabilities in the Context of the Right to Be Forgotten. In *Conference on Information and Knowledge Management* (*CIKM*), 2025.

[9] Xu Cao, Yifan Shen, Bolin Lai, **Wenqian Ye**, Yunsheng Ma, Joerg Heintz, Jintai Chen, Jianguo Cao, and James M Rehg. What is the Visual Cognition Gap between Humans and Multimodal LLMs? In *Conference on Language Modeling* (*COLM*), 2025.

[10] Can Cui, Zichong Yang, Yupeng Zhou, Juntong Peng, Sung-Yeon Park, Cong Zhang, Yunsheng Ma, Xu Cao, **Wenqian Ye**, Yiheng Feng, Jitesh H. Panchal, Lingxi Li, Yaobin Chen, and Ziran Wang. On-Board Vision-Language Models (VLMs) for Personalized Motion Control of Autonomous Vehicles. In *IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*), 2025.

[11] Guangtao Zheng, **Wenqian Ye**, and Aidong Zhang. Benchmarking Spurious Bias in Few-Shot Image Classifiers. In *European Conference on Computer Vision* (*ECCV*), 2024.

[12] Guangtao Zheng, **Wenqian Ye**, and Aidong Zhang. Spuriousness-Aware Meta-Learning for Learning Robust Classifiers. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (*KDD*), 2024.

[13] Guangtao Zheng, **Wenqian Ye**, and Aidong Zhang. Learning Robust Classifiers with Self-Guided Spurious Correlation Mitigation. In *International Joint Conference on Artificial Intelligence* (*IJCAI*), 2024.

[14] Yunsheng Ma, Xu Cao, **Wenqian Ye**, Can Cui, Kai Mei, and Ziran Wang. Learning Autonomous Driving Tasks via Human Feedbacks with Large Language Models. In *Findings of the Conference on Empirical Methods in Natural Language Processing* (*EMNLP Findings*), 2024.

[15] Xu Cao, Tong Zhou, Yunsheng Ma, **Wenqian Ye**, Can Cui, Kun Tang, Zhipeng Cao, Kaizhao Liang, Ziran Wang, James Rehg, and Chao Zheng. MAPLM: A Real-World Large-Scale Vision-Language Dataset for Map and Traffic Scene Understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2024.

[16] Yunsheng Ma, Can Cui, Xu Cao, **Wenqian Ye**, Peiran Liu, Juanwu Lu, Amr Abdelraouf, Rohit Gupta, Kyungtae Han, Aniket Bera, James Rehg, and Ziran Wang. LaMPilot: An Open Benchmark Dataset for Autonomous Driving with Language Model Programs. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2024.

[17] Xu Cao†, **Wenqian Ye**†, Kenny Moise, and Megan Coffee. MpoxVLM: A Vision-Language Model for Diagnosing Skin Lesions from Mpox Virus Infection. In *Machine Learning for Health Symposium* (*ML4H*), 2024.

[18] Adibvafa Fallahpour, Mahshid Alinoori, **Wenqian Ye**, Xu Cao, Arash Afkanpour, and Amrit Krishnan. EHRMamba: Towards Generalizable and Scalable Foundation Models for Electronic Health Records. In *Machine Learning for Health Symposium* (*ML4H*), 2024.

[19] **Wenqian Ye**, Yunsheng Ma, Xu Cao, and Kun Tang. Mitigating Transformer Overconfidence via Lipschitz Regularization. In *Conference on Uncertainty in Artificial Intelligence* (*UAI*), 2023.

[20] Xu Cao†, **Wenqian Ye**†, Elena Sizikova, Xue Bai, Megan Coffee, Hongwu Zeng, and Jianguo Cao. ViTASD: Robust ViT Baselines for Autism Spectrum Disorder Facial Detection. In *IEEE International Conference on Acoustics, Speech and Signal Processing* (*ICASSP*), 2023.

[21] Yunsheng Ma, **Wenqian Ye**, Xu Cao, Amr Abdelraouf, Kyungtae Han, Rohit Gupta, and Ziran Wang. CEMFormer: Learning to Predict Driver Intentions from In-Cabin and External Cameras via Spatial-Temporal Transformers. In *IEEE International Conference on Intelligent Transportation Systems* (*ITSC*), 2023.

Peer-Reviewed Workshop Publications

[22] Can Cui†, Yunsheng Ma†, Xu Cao†, **Wenqian Ye**†, Yang Zhou, Kaizhao Liang, Jintai Chen, Juanwu Lu, Zichong Yang, Kuei-Da Liao, Tianren Gao, Erlong Li, Kun Tang, Zhipeng Cao, Tong Zhou, Ao Liu, Xinrui Yan, Shuqi Mei, Jianguo Cao, Ziran Wang, and Chao Zheng. A Survey on Multimodal Large Language Models for Autonomous Driving. In *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops* (*WACVW*), 2024.

[23] **Wenqian Ye**†, Yunsheng Ma†, and Xu Cao. Uncertainty Estimation in Deterministic Vision Transformer. In *AAAI Workshop on Uncertainty Reasoning and Quantification in Decision Making*, 2023.

Work Experience

2026/05 – 2026/08 **Meta AI**
Research Scientist Intern
Research on Ranking AI and Model Intelligence for Generative Recommendation Systems.

2025/05 – 2025/08 **Amazon AWS AI**
Applied Scientist Intern
Team of Agentic AI, Fundamental Research. Research on improving agentic alignment with reward reasoning models.

2023/08 – Present **University of Virginia** Charlottesville, VA
Graduate Research Assistant
Conduct research on improving robustness and alignment of machine learning models. Publish research in top AI and data mining conferences, and develop open-source tools.

2022/01 – Present **NYU Langone Health** New York, NY
Adjunct Researcher
Conduct part-time research on Artificial Intelligence-enabled diagnosis of Tuberculosis and COVID-19 using radiologic imaging in resource-constrained environments. Lead the development of AI algorithmic frameworks (e.g., VLMs and AI agents) for screening Monkeypox using dermatologic images.

2022/07 – 2023/05 **Cirrus Logic Inc** Austin, TX
Software Engineer
Performed comprehensive validation and testing of embedded software for audio and haptics applications, focusing on automation and analysis. Contributed to both internal and customer-facing UI design, while executing system-level testing across device drivers, firmware, and UI software. Developed prototypes of DSP algorithms using Python/Matlab and implemented fixed-point firmware using C/C++.

Honors and Awards

2026 John A. Stankovic Graduate Research Award

2026 Tinker Teaching Grant (CS 6501 Workshop on Building AI Agents)

2025 NeurIPS Scholar Award

2025 **KDD Best Paper Award (Research Track)** (1 out of ∼3000 submissions)

2024 OpenAI Researcher Access Program ($5000)

2023 UAI Travel Award

2023 AAAI Travel Award

2023 UVA Computer Science Fellowship

Teaching Experience

Teaching Assistant

Spring 2026 **CS 6501 Workshop on Building AI Agents** University of Virginia
Instructor: Prof. Henry Kautz

Fall 2025	**CS 4774: Machine Learning** *Instructor: Prof. Hadi Daneshmand*	University of Virginia
Spring 2025	**CS 6501/4501: Analyzing Online Behavior for Public Health** *Instructor: Prof. Henry Kautz*	University of Virginia
Fall 2024	**CS 4501: Natural Language Processing** *Instructor: Prof. Yu Meng*	University of Virginia
Fall 2021	**CSCI-GA 2590: Natural Language Processing** *Instructor: Prof. He He*	New York University

Guest Lecturer

Fall 2024	**CS 6316: Machine Learning** *Instructor: Prof. Aidong Zhang* Topic: Spurious Correlations in Machine Learning	University of Virginia

Selected Invited Talks

2026/03 **From Spurious Correlations to Emergent Misalignment: A Path Toward More Reliable AI**
Microsoft Copilot Tuning

2026/02 **From Spurious Correlations to Emergent Misalignment: A Path Toward More Reliable AI**
National Science Foundation (NSF) PPoSS Meeting

2025/08 **Improving Inter-Agent Alignment with Agentic Reward Reasoning**
Amazon AWS AI

Services

Organizer

Co-organizer/Program Chair
WDFM-EAI Workshop (CVPR 2026);
WDFM-AD Workshop (CVPR 2025; ICCV 2025);
LLVM-AD Workshop (WACV 2024; ITSC 2024; WACV 2025);
AI4CHL Workshop (ICLR 2025);

Roundtable Chair
ML4H 2024

Program Committee

Journals
ACM TIST; IEEE TPAMI; IEEE IoT-J; IJHCI; IEEE T-IV; IEEE VTM; IEEE Internet Computing

Conferences
ICML; ICLR; NeurIPS; KDD; CVPR; ECCV; ICCV; AAAI; IJCAI; AISTATS; ICASSP; MICCAI; ISBI; ACML

Membership

Member
IEEE; ACM; AAAI; IEEE SPS

Mentorship

Mentorship Chair
UVA CS Graduate Student Group (CSGSG)

Mentor
ML4H(2023, 2024)

Technical Skills

Languages	**Python**, C/C++, R, MATLAB, Golang, SystemVerilog, LaTeX
Packages	PyTorch, TensorFlow, AG2, LangChain/LangGraph, Huggingface, Scikit-learn
Others	AWS Bedrock, CUDA, SQL, Git, Jenkins